

Sustainable Reaction Engineering Systems: A Literature Review

Part 1

Sustaining human life beyond Earth requires closing water, oxygen, and carbon loops under severe mass, power, and reliability constraints. Unlike terrestrial chemical plants, space-based systems must operate in extreme environments with strict limits on volume, energy consumption, and maintenance, while running autonomously for extended periods without resupply from Earth. The ISS Environmental Control and Life Support System (ECLSS) provides the most mature demonstration of closed-loop life support. It integrates water recovery, air revitalization, and oxygen generation through catalytic and electrochemical processes. Mars in-situ resource utilization (ISRU) systems will face even greater demands, requiring conversion of the Martian atmosphere into propellants, oxygen, and water.

This review examines the catalysts, reactors, and integration strategies that enable closed-loop life support, with emphasis on the Sabatier reaction and PEM electrolysis. Future improvement pathways are discussed, followed by an assessment of artificial intelligence applications in chemical reaction engineering.

The ISS ECLSS has three primary subsystems. The Water Recovery System reclaims approximately 90% of station water from urine, humidity condensate, and suit hydration systems. The Air Revitalization System removes CO₂ and trace contaminants using molecular sieves, activated charcoal, and catalytic oxidizers. The Oxygen Generation System produces O₂ via water electrolysis and recovers oxygen from metabolic CO₂ through the Sabatier reaction. The Oxygen Generation Assembly (OGA) uses a PEM water electrolyzer to produce oxygen for crew respiration, accounting for roughly 33% of ECLSS power consumption. The current ISS OGA uses a cathode-feed configuration. Operating conditions include a cathode inlet water flow of 8.07 kg/hr at 339 K, average stack temperature of 335 K, and system pressure of 2860 kN/m² (approximately 28 atm). The stack operates at 75 A and approximately 1.72 V per cell, producing 2.3 to 9.2 kg of oxygen per day. (Akay et al., 2025)

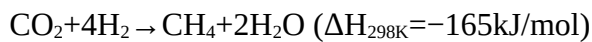
Microgravity presents a specific challenge for PEM electrolysis: gas bubbles do not detach from electrode surfaces as they would under buoyancy on Earth. Instead, bubbles accumulate between the membrane and electrode, increasing mass transport resistance and reducing efficiency. This requires careful flow field and separator design to maintain wetting and promote bubble removal. Collins Aerospace has proposed a NextGen OGA using anode-feed commercial stacks, which achieve current densities of 2-5 A/cm² versus the legacy system's <0.5 A/cm². At 0.7 A/cm², the system can support a 10-person crew. However, this introduces two-phase flow challenges. Orientation testing by Myles et al. (2024) found that an upside-down configuration at 4 LPM caused voltage spread to increase from 35 to 100 mV over 300 hours due to gas blanketing and membrane dry-out. Increasing flow to 5 LPM restored stability, indicating that microgravity operation requires active flow management.

The anode-feed configuration also permits a quantitative approach to hydrogen safety. The legacy system used triple seals and a containment dome. The NextGen approach compares maximum credible leak (MCL) to maximum allowable leak (MAL). With 65 psia H₂ pressure, dissolved H₂ in

O₂ remains below 0.1%. Phase separator efficiency above 99.5% maintains O₂ product below 1% H₂ (25% of the lower flammability limit), enabling elimination of the containment dome (Myles et al., 2024).

Roy et al. (2019) investigated PEM-based oxygen concentration and compression. Their Solid-State Oxygen Concentrator and Compressor (SSOCC) operates as a combined electrolyzer and fuel cell, extracting O₂ from cabin air and compressing it to 3600 psi with no moving parts. At 100 mA/cm² and 35°C, the SSOCC achieved 270 hours of operation at 0.8V.

The Sabatier reaction converts metabolic CO₂ and electrolytic hydrogen into methane and water:



This exothermic reaction is reversible. Lower temperatures favor equilibrium conversion, but kinetics require elevated temperatures for reasonable rates.

The Hamilton Sundstrand reactor on the ISS operates at approximately 515°C and ambient pressure. CO₂ enters at 48°C. Once initiated, the reaction is thermally self-sustaining, achieving >99% CO₂ conversion. The CO₂ Reduction Assembly launched in 2009 and became operational in June 2011.

The catalyst is proprietary ruthenium/alumina (Ru/Al₂O₃). Ruthenium offers higher activity and selectivity than nickel, though at greater cost. Other effective catalysts include cobalt and rhodium.

Several issues have emerged with the ISS Sabatier reactor. The catalyst eventually deactivated, leading to reactor shutdown. The specific poison has not been publicly identified, but trace volatile organic compounds in the feed stream are suspected. High temperatures shift equilibrium toward reactants. Increasing CO₂ pressure can shift the balance back toward products. Methane is vented because storing flammable gas on the ISS is hazardous. This loses hydrogen permanently, limiting theoretical oxygen recovery from CO₂ to approximately 54%. Current recovery is around 42%. Gas-liquid separation is complicated in microgravity, and particulate formation can occur.

The three subsystems must be integrated for overall efficiency. The mass balance loop operates as follows:

1. Crew consumes O₂ and produces CO₂ (~1 kg/person-day)
2. PEM electrolyzer splits water into O₂ and H₂
3. Sabatier reactor combines H₂ with exhaled CO₂ to produce CH₄ and H₂O
4. Water is recycled to the electrolyzer and for crew consumption
5. CH₄ is vented

NASA has identified the need for >75% oxygen recovery for future missions. This requires technologies such as methane pyrolysis or plasma pyrolysis to recover hydrogen from the methane stream.

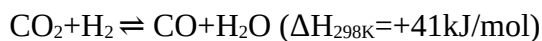
For a Martian colony, reaction engineering must support not only life support but also propellant production, materials manufacturing, and a self-sustaining chemical economy. The Martian atmosphere provides a carbon and oxygen source, but extraction and conversion require sophisticated engineering.

NASA's ISRU program is developing technologies to convert Martian CO₂ and water into propellants (methane and oxygen), reducing the mass of payloads launched from Earth.

The Sabatier reaction is central to Mars methane production. The Atmospheric Processing Module (APM), an ISRU technology demonstrator, used a packed bed reactor with Ru/Al₂O₃ catalyst pellets. CO₂ was captured from simulated Martian atmosphere using cryocoolers, then sublimated and pressurized as feedstock.

APM testing revealed catalyst failure at high flow rates and elevated temperatures, detected through unwanted byproduct formation. This prompted development of computational models to predict acceptable operating conditions and avoid catalyst damage (Meier et al., n.d.). For a Mars mission, the catalyst must withstand variable conditions for up to 480 days to produce seven tons of methane.

While Sabatier produces methane and water, it does not directly produce oxygen. The Reverse Water-Gas Shift reaction provides an additional pathway:



This endothermic reaction competes with Sabatier in CO₂ hydrogenation and is favored at high temperatures. Its products can be further processed: CO as fuel or feedstock, H₂O electrolyzed to O₂.

Thermal integration presents an opportunity: excess heat from the exothermic Sabatier reaction can drive the endothermic RWGS process.

Pacific Northwest National Laboratory has developed microchemical and thermal systems for Mars ISRU, using channels on the order of 100 microns to enable rapid heat and mass transfer (TeGrotenhuis et al., n.d.). Advantages include:

- Orders of magnitude reduction in volume and mass
- Extensive heat recuperation with minimal size penalty
- High reliability through massive parallelization
- Precise control of reaction conditions

Catalyst development for microchannel reactors has focused on Sabatier and RWGS reactions. Hu et al. (2007) found that 3% Ru/TiO₂ and 6% Ru/CeO₂-ZrO₂ were active Sabatier and RWGS catalysts, achieving near-equilibrium conversion at high throughputs. In stability testing under repeated start-up and shut-down cycles, both catalysts retained initial activities.

Solid Oxide Electrolysis Cells (SOEC) operate at 700-1000°C, where thermal energy partially replaces electrical energy in driving CO₂ decomposition:

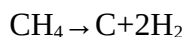


This direct conversion offers oxygen production without the hydrogen requirement of Sabatier/RWGS systems. The Mars Oxygen In Situ Resource Utilization Experiment (MOXIE), deployed on the Perseverance rover, demonstrated oxygen production by high-temperature solid-oxide electrolysis of atmospheric CO₂. OxEon Energy delivered the SOEC stack for MOXIE, which produced propellant-grade oxygen by electrolyzing Mars atmosphere CO₂.

Current ISS systems rely on molecular sieves and lithium hydroxide beds. Metal-Organic Frameworks (MOFs) offer potential for higher selectivity, greater stability, and poison resistance. MOFs can be chemically functionalized for molecule-specific adsorption, potentially enabling more efficient CO₂ concentration before the Sabatier reaction.

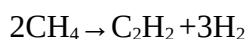
Electrochemical CO₂ reduction (ECO₂R) is an increasingly valuable technology that converts CO₂ into useful chemicals using various catalyst materials. Low-temperature electrolysis (25-80°C) offers advantages over high-temperature methods for space applications, particularly for Mars ISRU (Govea-Alvarez et al., 2026).

Methane pyrolysis can recover hydrogen from methane:



The recovered hydrogen can be recycled to the Sabatier reactor, while carbon byproduct could be used for radiation shielding or water purification. Childers et al. (n.d.) from Honeywell Aerospace demonstrated 50-80% conversion with >95% selectivity. Methane conversion increases with higher temperatures and decreases with shorter residence times. Hydrogen recovery is extremely valuable in space because it can be used as propellant for rocket ships on Mars.

Plasma pyrolysis offers an alternative route:



Wheeler et al. (n.d.) demonstrated the Plasma Pyrolysis Assembly (PPA) under zero-g conditions on a parabolic flight. The reactor uses a microwave to ignite a hydrogen-methane plasma at approximately 385 °C. Parabolic flight experiments confirmed the PPA's ability to transition from H₂ to H₂/CH₄ plasma under zero-g and ignite H₂ plasma in Martian and Lunar gravities. However, practical challenges include power consumption, size and weight constraints, and maintenance requirements.

Key catalyst development priorities include:

1. Durability: Catalysts must withstand years of continuous operation
2. Poison resistance: Protection against trace contaminants (VOC)
3. Lower temperature operation: Reducing energy requirements
4. Reduced precious metal content: Lowering cost and supply chain dependence

Research into supported ruthenium catalysts for Mars missions is ongoing, with focus on understanding catalyst life and performance issues expected on Mars. The effects of launch vibration loads on catalyst integrity must also be considered.

Energy efficiency is critical for space-based chemical processing. Future systems will require:

1. Heat recuperation between exothermic and endothermic processes
2. Energy cascading (high-grade heat for reactions, low-grade heat for other uses)
3. Micro-channel heat exchangers for efficient transfer with minimal mass penalty

The low-pressure Martian environment makes convective heat transfer ineffective, so radiators must be carefully designed.

Materials compatibility in high-pressure oxygen environments presents constraints often overlooked. Roy et al. (2019) report that CP titanium is flammable at 5 psia O₂ and CRES 316 at 150 psia. Only Haynes 25, Hastelloy C276, and Inconel 625 have flammability thresholds of 2500 psia. Even these require composite structures due to pitting corrosion under anodic potential.

First, methane pyrolysis deserves greater investment. The current oxygen recovery ceiling of 42-54% is the single biggest inefficiency in the ISS ECLSS. While plasma pyrolysis has been demonstrated in zero-g, the practical challenges of power consumption and carbon handling remain unresolved. A simpler thermal pyrolysis reactor with integrated carbon capture would be preferable. Honeywell's 50-80% conversion is promising, but the technology needs to move from demonstration to flight qualification. For Mars missions, recovering hydrogen from methane is essential.

Second, the NextGen OGA's anode-feed architecture is the right direction. Commercial PEM stacks offer proven reliability and higher current densities than the legacy cathode-feed system. However, the orientation testing results from Myles et al. (2024) show that microgravity operation requires careful flow management. I suspect that pulsed flow or variable flow control could improve gas removal in microgravity beyond the steady-state 5 LPM solution. This is an area where active control could prevent the gas blanketing observed in testing.

Third, material compatibility in high-pressure oxygen is an underappreciated constraint. The flammability data from Roy et al. (2019) should drive early-stage design decisions. Many catalyst and electrode development efforts focus on electrochemical performance while assuming that materials qualified for terrestrial use will translate to space. This is not a safe assumption. The acidic, oxidizing environment of a PEM electrolyzer anode is uniquely harsh, and oxygen compatibility testing under wet conditions (as proposed by Collins Aerospace) should be a prerequisite for material selection.

Finally, the most impactful improvement may be system-level rather than component-level. Thermal integration between Sabatier and RWGS is an obvious win that has been demonstrated at lab scale but not yet integrated into flight hardware. Similarly, the trade-off between ECLSS and ISRU thermal management on Mars deserves more analysis. Preliminary studies suggest that co-located units could reduce radiator mass by 5-15% (Akay et al., 2025). This kind of integration may offer larger gains than any single catalyst improvement.

Part 2. The Impact of Artificial Intelligence on Chemical Reaction Engineering and Analysis

AI has attracted significant attention in chemical reaction engineering, with applications ranging from catalyst screening to autonomous reactor control. For space systems, autonomous optimization and fault diagnosis is very valuable.

Traditional catalyst discovery relies on experimental trial-and-error or computationally expensive density functional theory (DFT) calculations. Machine learning models trained on existing data can predict catalyst activity, selectivity, and stability.

Wang et al. (2025) at Tsinghua University developed the CaTS framework using transfer learning, requiring only hundreds of catalytic reaction data points to train machine learning force fields. This approach improves transition-state search efficiency by nearly 10,000 times while maintaining

consistency with DFT. The DigMethpy platform combines literature, experimental data, simulations, and ML models into a unified discovery framework.

Recent work has demonstrated fine-tuning large language models to generate single-atom catalyst synthesis procedures. Given that heterogeneous catalyst design is primarily expert-driven and literature is growing rapidly, LLMs could help capture and disseminate synthesis knowledge.

The Reac-Discovery platform integrates reactor design, fabrication, and optimization based on machine learning, enabling parallel multi-reactor evaluations with real-time NMR monitoring.

Self-optimizing continuous-flow systems integrate machine learning and robotics, autonomously exploring parameter spaces and adapting to changing conditions. These capabilities are invaluable for space applications where remote operation is essential.

Machine learning force fields trained on DFT data can achieve quantum-mechanical accuracy at reduced computational cost, enabling simulation of complex catalytic systems and providing insights into reaction mechanisms.

AI can accelerate kinetic model development by automatically selecting appropriate rate expressions and estimating parameters from experimental data. For the Sabatier reaction, computational models already predict temperature profiles and identify acceptable operating conditions to avoid catalyst damage (Meier et al., n.d.). AI-enhanced models could further improve these predictions. AI can assist in transition state searches by searching the entire reaction space for the minimum energy path.

Machine learning models integrated with sensors enable real-time optimization. For PEM electrolyzers, AI could optimize current density, temperature, and pressure to maximize O₂ production while minimizing energy consumption.

Predictive maintenance models can forecast catalyst deactivation and system failures, enabling preventive action. AI systems can also analyze sensor data to diagnose faults and recommend corrective actions, reducing the need for ground intervention.

A Mars-based Sabatier reactor could incorporate AI for autonomous operation, monitoring catalyst activity through product composition, adjusting temperature and feed ratios to maintain conversion, predicting catalyst regeneration needs, and integrating with RWGS and electrolysis processes. AI could accelerate development of non-precious metal catalysts for PEM electrolysis by screening thousands of compositions computationally, predicting activity and stability, guiding synthesis, and iteratively refining predictions based on experimental feedback. AI could optimize microchannel geometry, catalyst placement, and operating conditions for Mars ISRU reactors, predicting heat and mass transfer characteristics to design compact, efficient reactors.

Several limitations remain. Despite this promise, the literature reveals a glaring gap: almost all these AI models are trained on 1g terrestrial data. Whether a force field trained on Earth-based bubble dynamics can predict gas blanketing in 0.38g Martian gravity is an open question. Furthermore, the Sabatier reactor models by Meier et al. are empirical; integrating a physics-informed neural network would require kinetic data that simply doesn't exist for long-term microgravity operation. Until we generate that dataset, AI will remain a design assistant rather than an autonomous controller.

AI offers potential for accelerating catalyst discovery, enabling autonomous reactor operation, and providing insights into reaction mechanisms. For space applications, where autonomous operation

is essential, AI could be particularly valuable. However, data scarcity and validation challenges limit current applicability. Physics-informed hybrid models, combining first-principles kinetics and machine learning may be the most practical path forward for catalyst design.

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